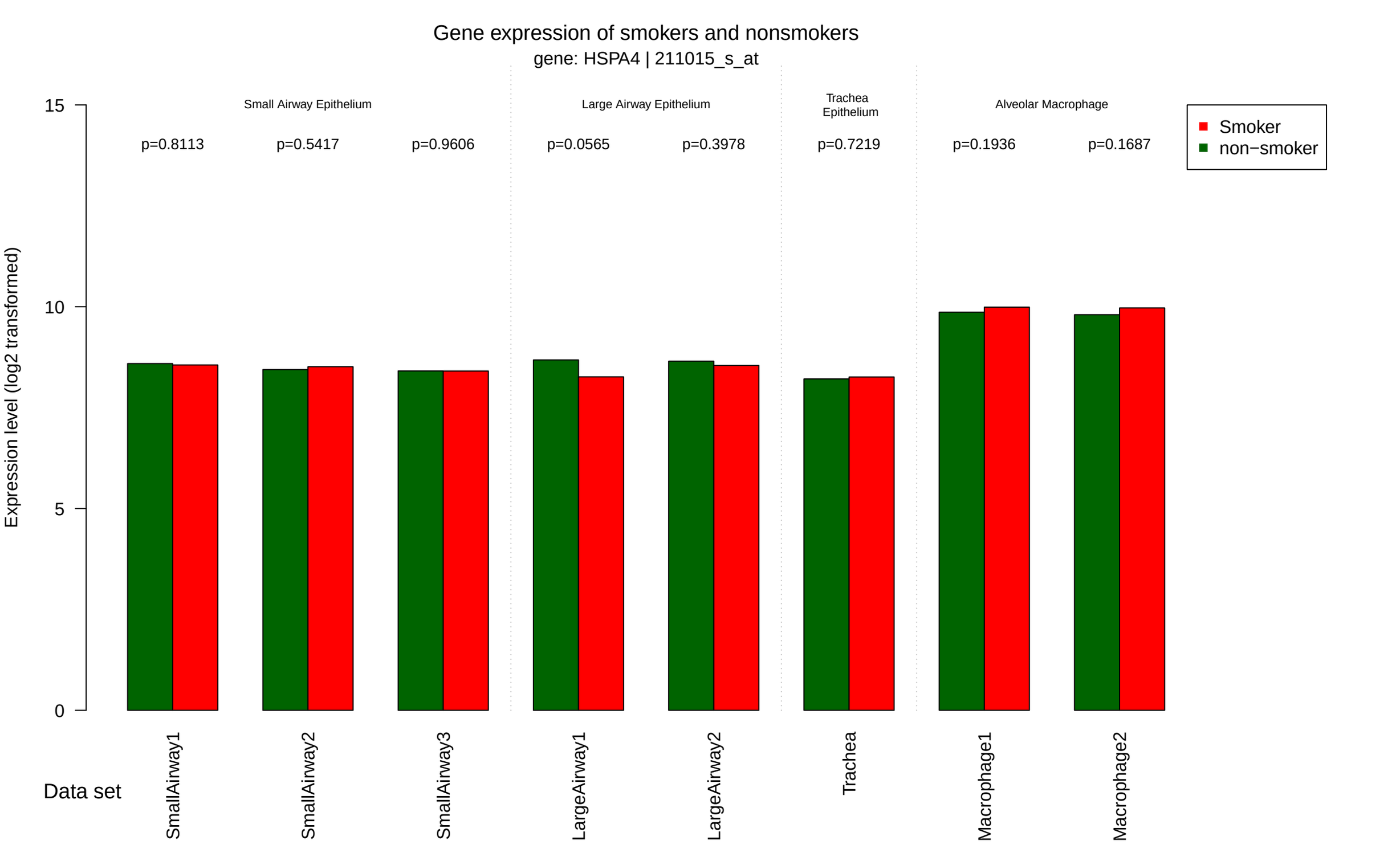
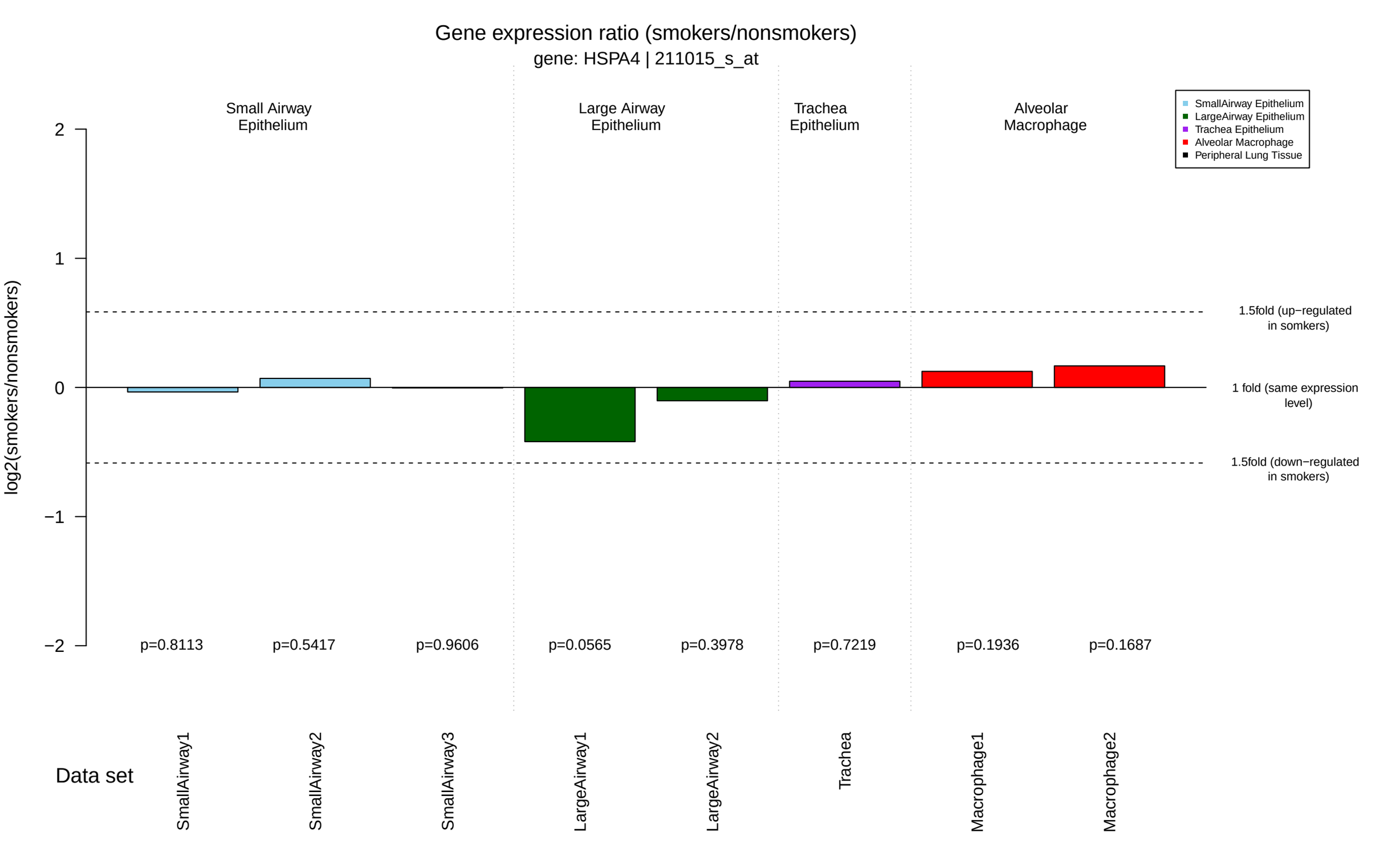
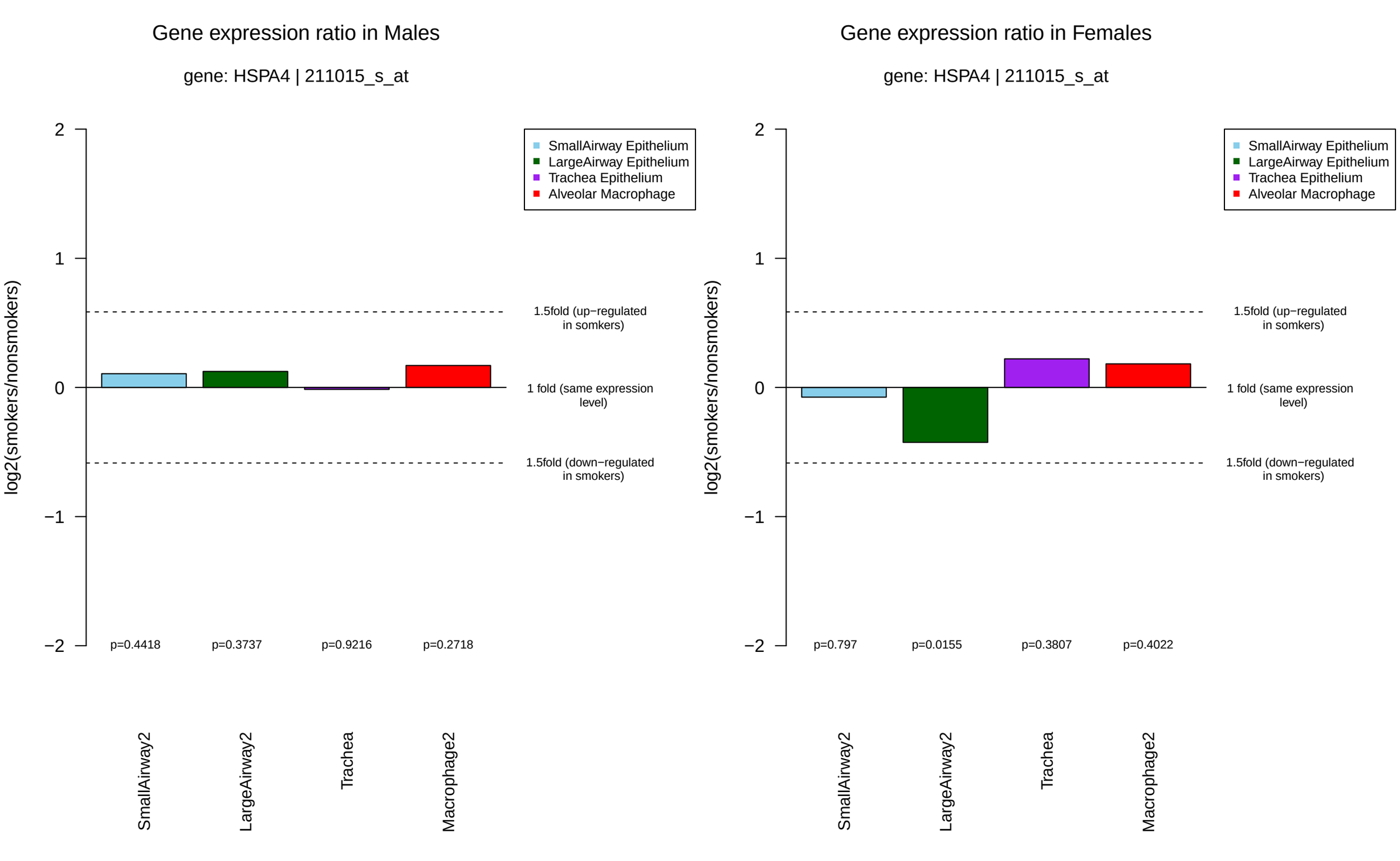
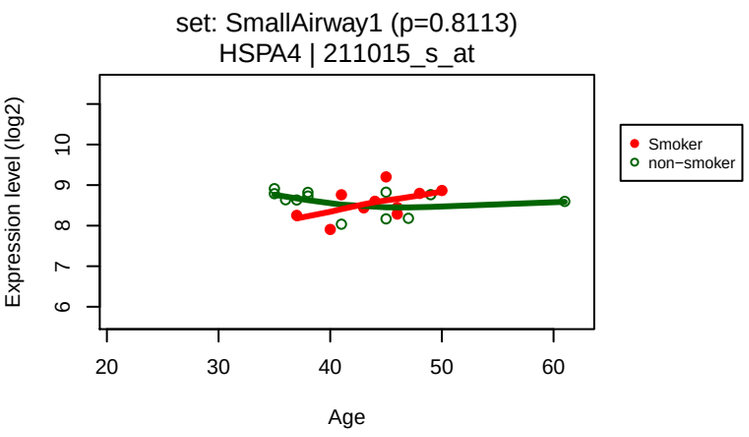
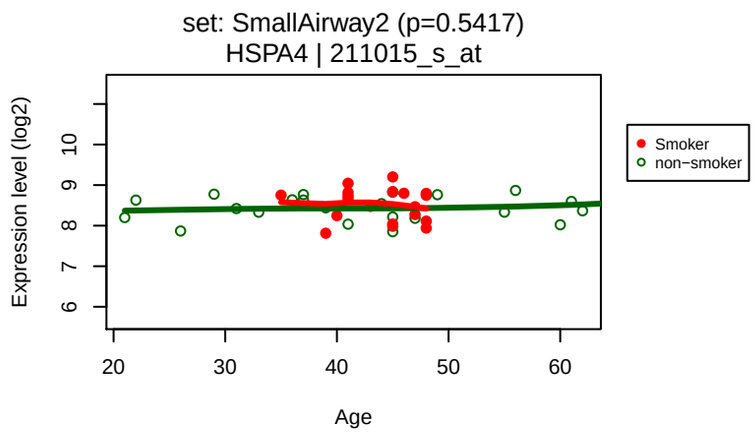
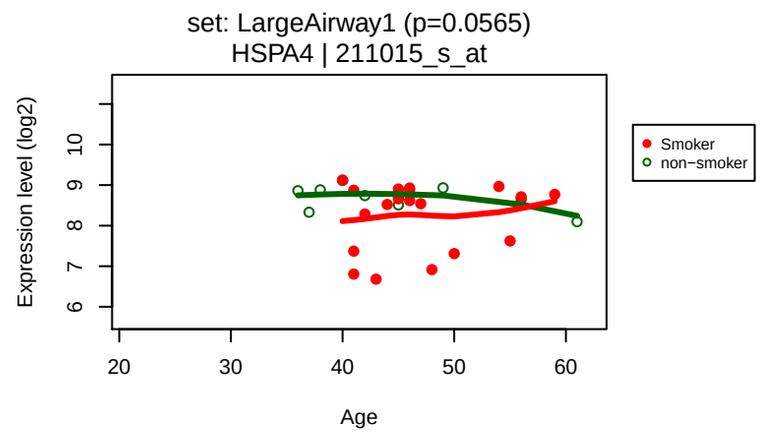

Gene expression of smokers and nonsmokers
gene: HSPA4 | 211015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8113
p=0.5417
p=0.9606
p=0.0565
p=0.3978
p=0.7219
p=0.1936
p=0.1687
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

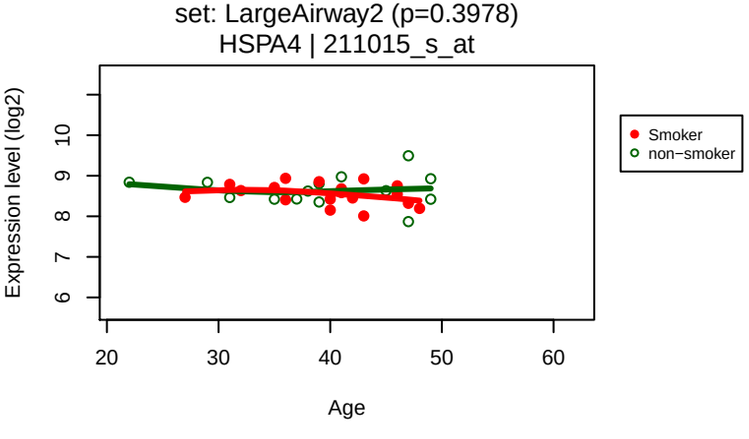

Gene expression ratio (smokers/nonsmokers)
gene: HSPA4 | 211015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8113
p=0.5417
p=0.9606
p=0.0565
p=0.3978
p=0.7219
p=0.1936
p=0.1687
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

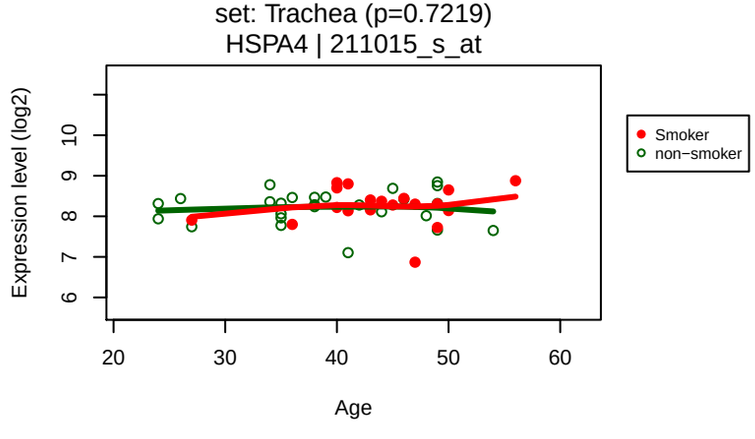

Gene expression ratio in Males
gene: HSPA4 | 211015_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4418
p=0.3737
p=0.9216
p=0.2718
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HSPA4 | 211015_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.797
p=0.0155
p=0.3807
p=0.4022
SmallAirway2
LargeAirway2
Trachea
Macrophage2

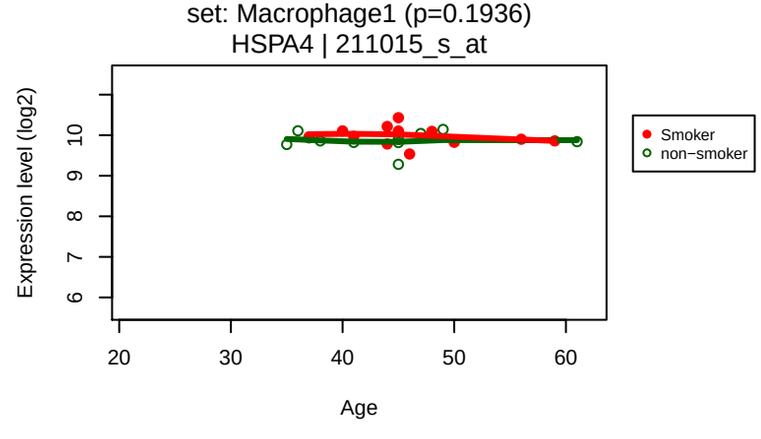

set: SmallAirway1 (p=0.8113)
HSPA4 | 211015_s_at
Expression level (log2)
Age
Smoker
non−smoker

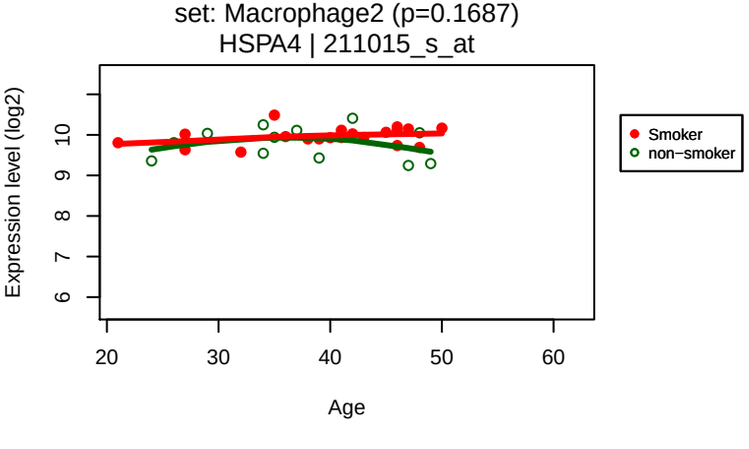

set: SmallAirway2 (p=0.5417)
HSPA4 | 211015_s_at
Expression level (log2)
Age
Smoker
non−smoker

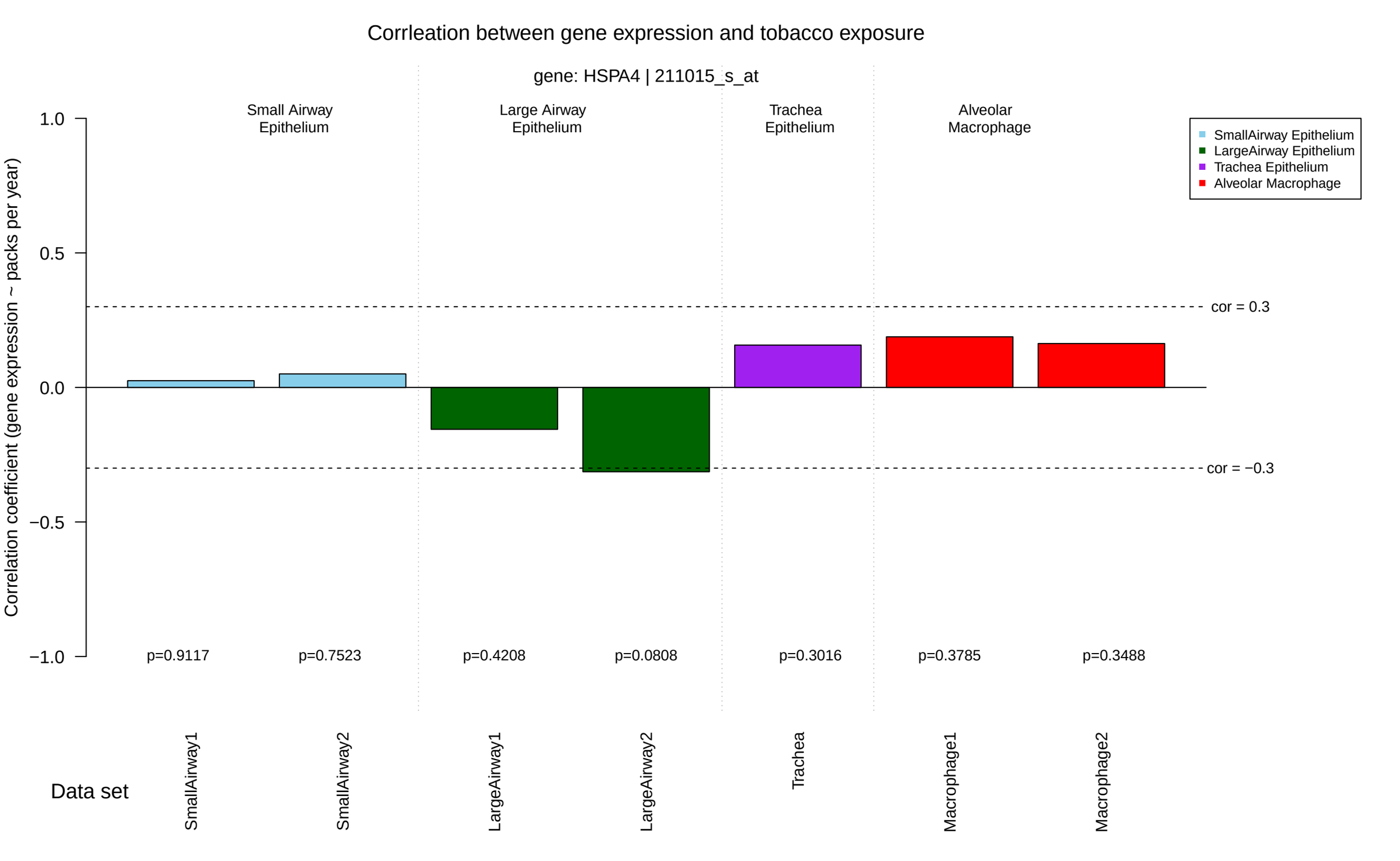

set: LargeAirway1 (p=0.0565)
HSPA4 | 211015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3978)
HSPA4 | 211015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7219)
HSPA4 | 211015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1936)
HSPA4 | 211015_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1687)
HSPA4 | 211015_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HSPA4 | 211015_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9117
p=0.7523
p=0.4208
p=0.0808
p=0.3016
p=0.3785
p=0.3488
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2